

AMENDMENT NO. 1 OCTOBER 1995
TO
IS 13508 : 1992 TEST PROCEDURE FOR MEASURE-
MENT OF LOSS TANGENT ANGLE OF COILS
AND BARS FOR MACHINE WINDING — GUIDE

(*Cover page and page 1, Title*) — Substitute the following for the existing title:

'TESTS PROCEDURE FOR MEASUREMENT
AND REQUIREMENTS OF LOSS TANGENT ANGLE OF
COILS AND BARS FOR MACHINE
WINDING — GUIDE'

(*Page 1, clause 3.6, line 2*) — Delete the word 'which'.

(*Page 2, clause 4.2, line 6*) — Substitute ' 100×10^{-3} ' for ' $100 \times 10^{\frac{1}{2}}$ '.

(*Page 2, clause 4.2, line 7*) — Substitute the word 'Sine' for 'cosine'.

[*Page 2, clause 5.1.2 (b)*] — Substitute 'Increment ($\tan \delta 0.6 U_N - \tan \delta 0.2 U_N$)' for 'Increment $\tan \delta 0.6 U_N - \tan \delta 0.2 U_N$ '.

(*Page 3, clause 5.2.2.1, line 3*) — Add the word 'the' after 'for' and delete the word 'the' after 'shielding'.

(*Page 4, clause 5.3*) — Add the following clauses after clause 5.3.1:

"5.4 Permissible Values of Loss Tangent

5.4.1 Values likely to be achieved on systems commonly used are indicated in 5.4.2.

5.4.2 The initial value of $\tan \delta 0.2 U_N$, the increment $0.5 \times (\tan \delta 0.6 U_N - \tan \delta 0.2 U_N)$ and the increment $\tan \delta$ per measuring step up to U_N shall not exceed the values in Table 1 for rated voltages up to 11 kV (*see also* Fig. 1). For rated voltages above 11 kV the values should be fixed by agreement between manufacturer and user.

5.4.3 If more than 5 percent of the samples show test results in the ranges between col 2 and 3 or between col 4 and 5 of Table 1, or in a different range of agreed values, the testing shall be continued with an equal number of further samples, if necessary up to the total number of bars or coil sides. If during these tests the values of Table 1, or the agreed values, are not exceeded, the test of insulation shall be regarded as satisfactory.

5.4.4 If an automatic measuring system is used, conformity to the standard shall be determined at the reference voltages defined in this standard.

Table 1 Highest Permissible Values of Loss Tangent
(Rated Voltage not Exceeding 11 kV)
(*Clauses 5.4.2 and 5.4.3*)

Tan Delta 0.2 U_N	Tan Delta - Tan Delta 0-6 U_N 0.2 U_N 2		Delta tan Delta per Step of 0.2 U_N	
(1)	(2)	(3)	(4)	(5)
All samples 30×10^{-3}	95 percent samples 2.5×10^{-3}	Remaining 5 percent samples 3×10^{-3}	95 percent samples 5×10^{-3}	Remaining 5 percent samples 6×10^{-3}

5.5 Sampling Plan

The recommended sampling plan for carrying out the test is given in Table 2.

Table 2 Number of Bars or Coil Sides to be Tested
(*Clause 5.5*)

No. of Poles	Rated Output	No. of Test Samples
All polarities	Less than 1 MVA or MW	5 percent of coil sides
All polarities	1 to 5 MVA or MW	10 percent of bars or coil sides with a minimum of 20
Two and four	50 MVA or MW or greater	All bars or coil sides
	Less than 50 MVA or MW but not less than 5 MVA or MW	At least 60 bars or 60 coil sides and in addition, 10 percent of all bars or coil sides
Six and more	5 MVA or MW or greater	

(ETD 4)